

Medicine Outcome Prediction System

SHAP versus LIME explainability comparison

Report scope

SHAP and LIME feature rankings are compared across the trained models. LIME one-hot encoded columns are aggregated back to their original clinical variables before comparison.

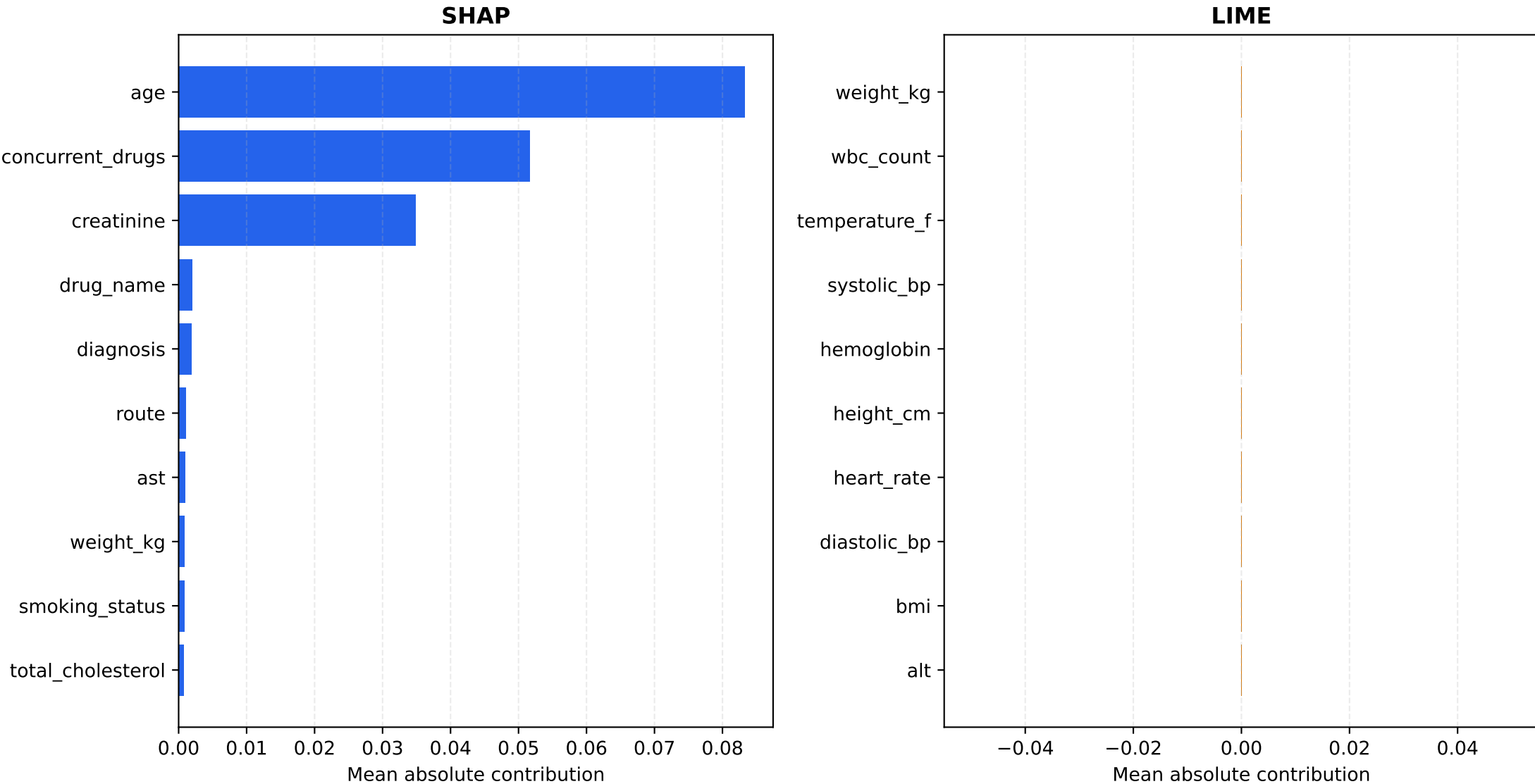
Interpretation

Top-10 overlap describes agreement between two model explanation methods; it does not establish clinical importance or causation. Zero LIME values are reported as an explanation limitation, not as proof that a model has no feature influence.

Models compared: 4

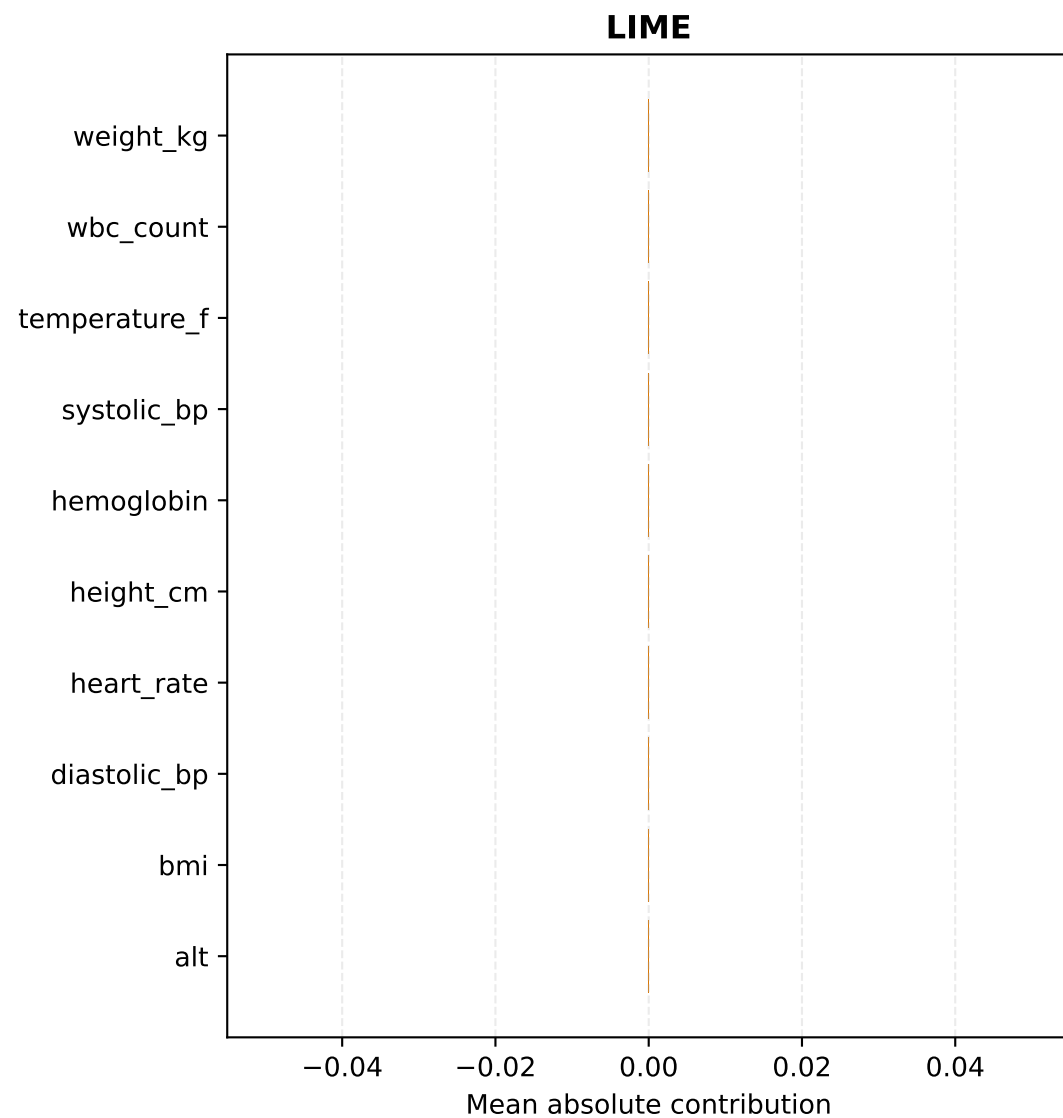
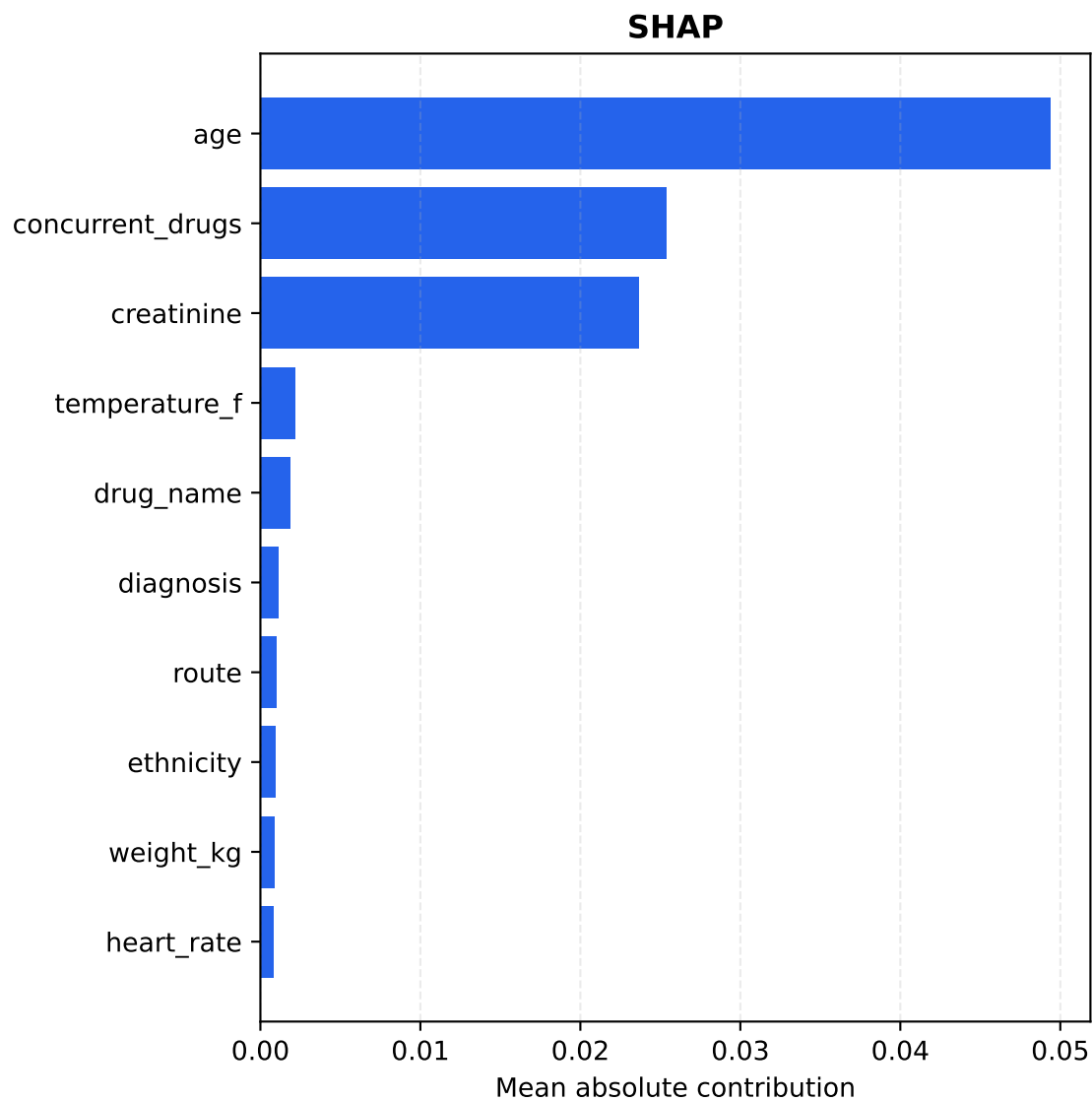
Methods: global SHAP importance and aggregated local LIME importance

Decision Tree: Top explainability features



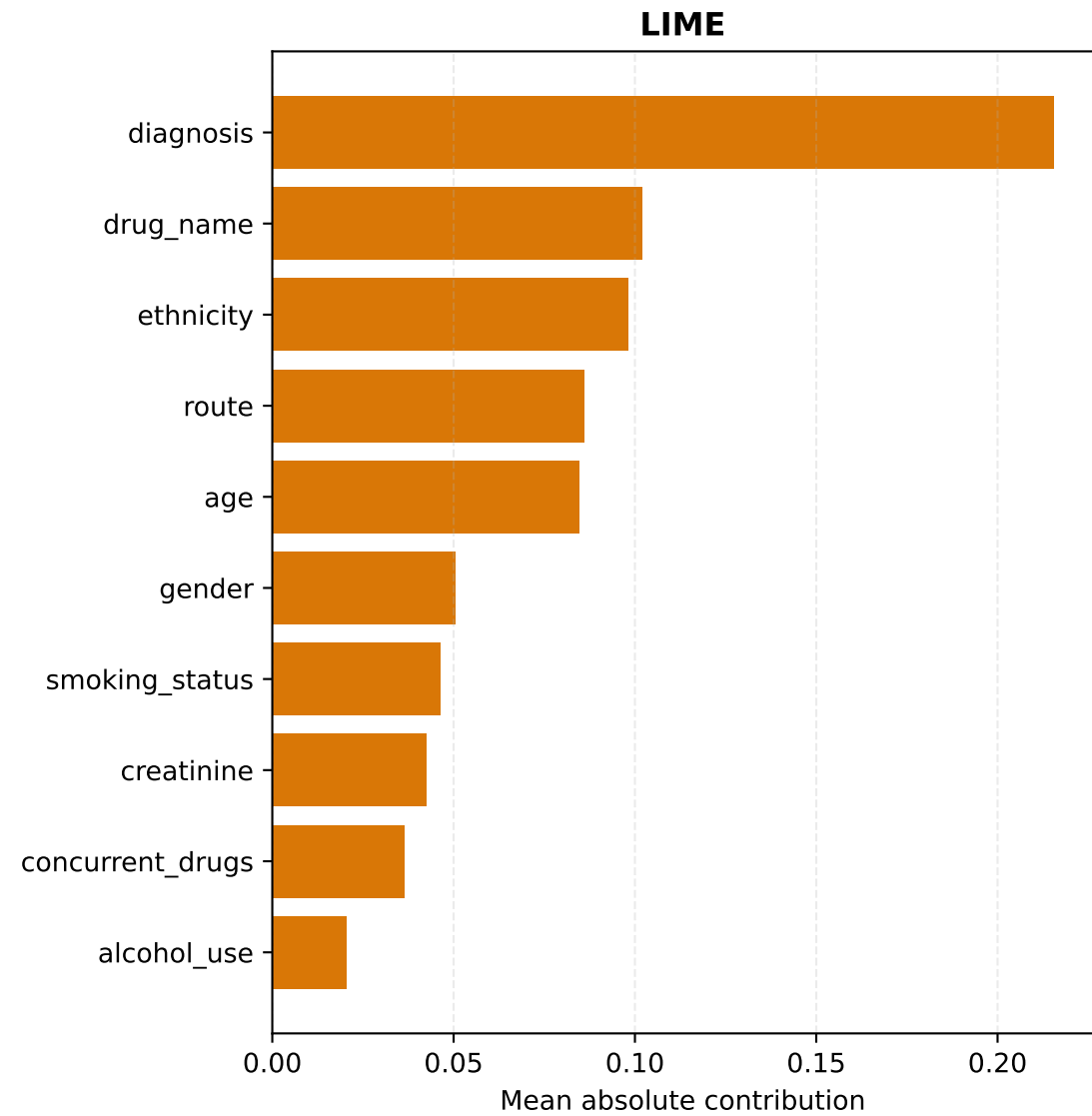
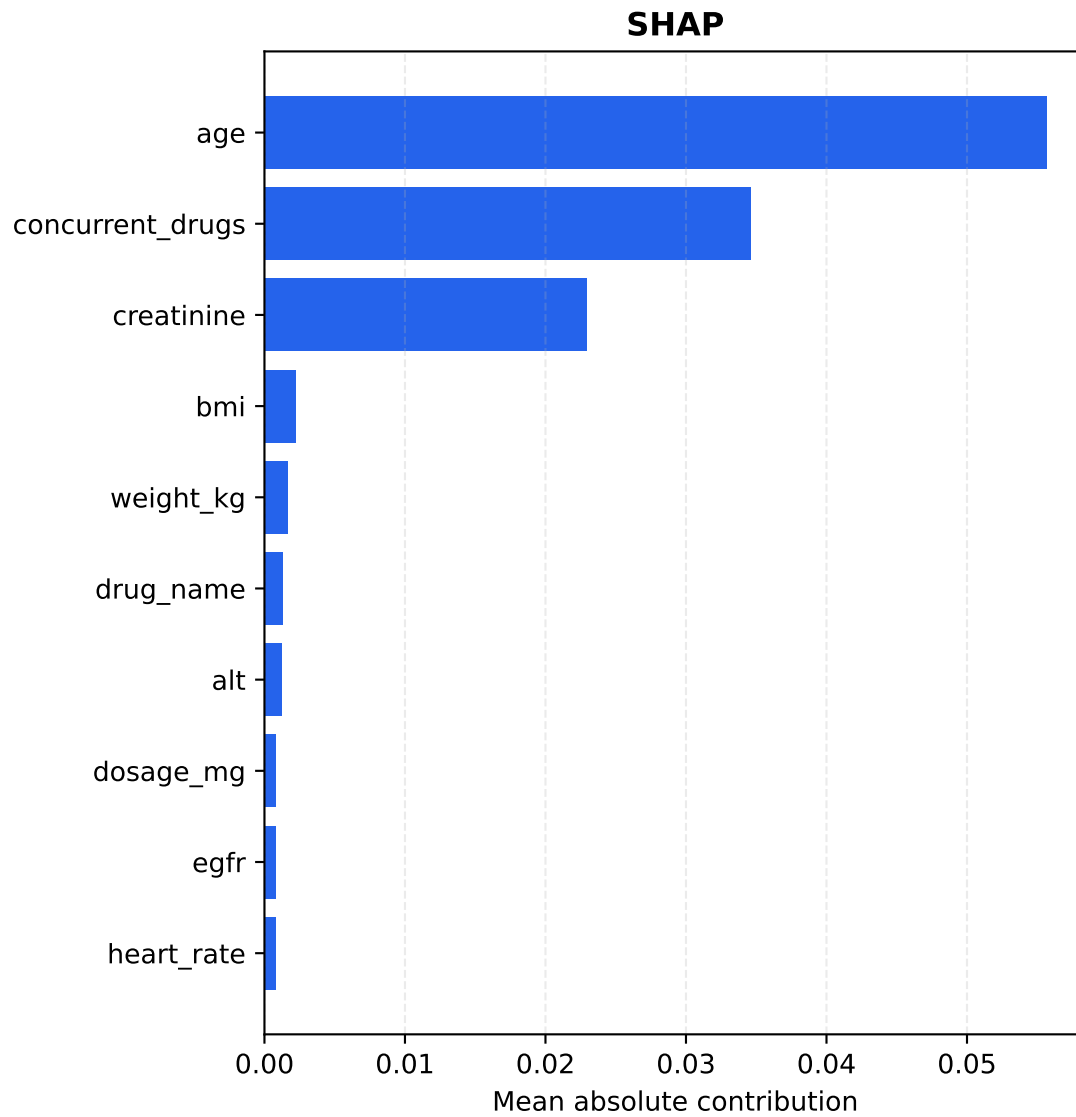
Top-10 overlap (1): weight_kg
LIME values are all zero in the available artifact.

Random Forest: Top explainability features



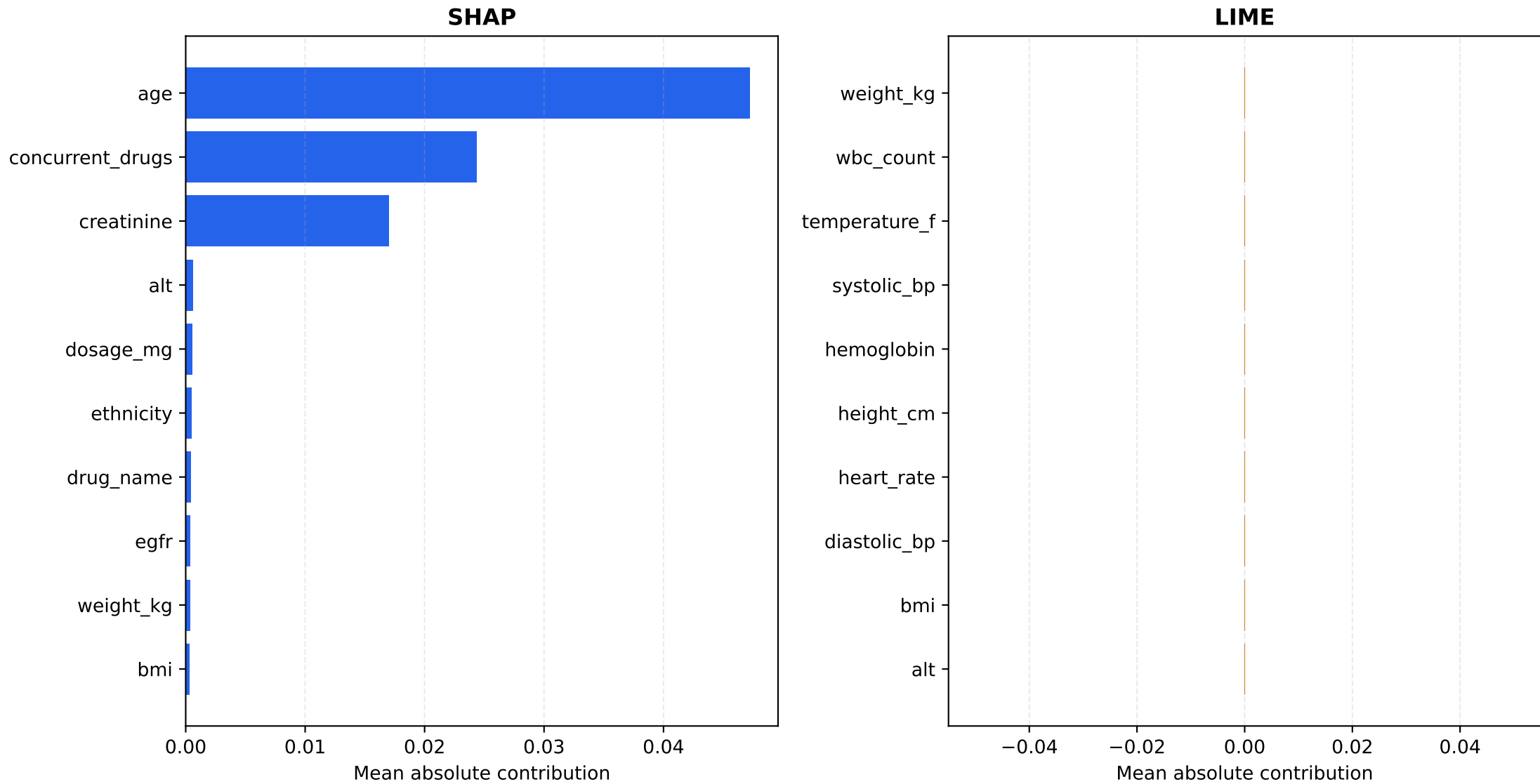
Top-10 overlap (3): temperature_f, weight_kg, heart_rate
LIME values are all zero in the available artifact.

XGBoost: Top explainability features



Top-10 overlap (4): age, concurrent_drugs, creatinine, drug_name

Gradient Boosting: Top explainability features



Top-10 overlap (3): alt, weight_kg, bmi
LIME values are all zero in the available artifact.